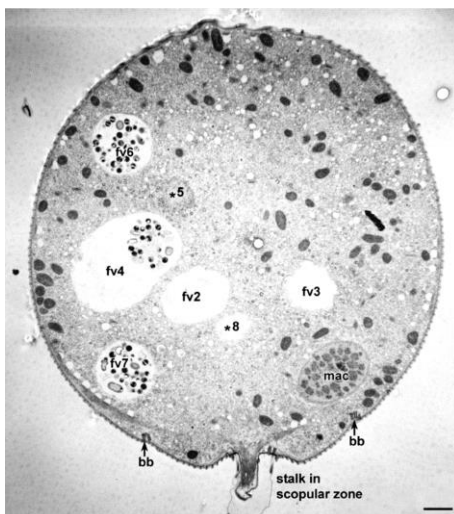


- 1 Vorticella is a genus of protozoa found mainly in freshwater streams. The transmission electron micrograph below shows food vacuoles (fv), a stalk (stalk) and a macronucleus (mac). The bar on the bottom right is 2 μ m.



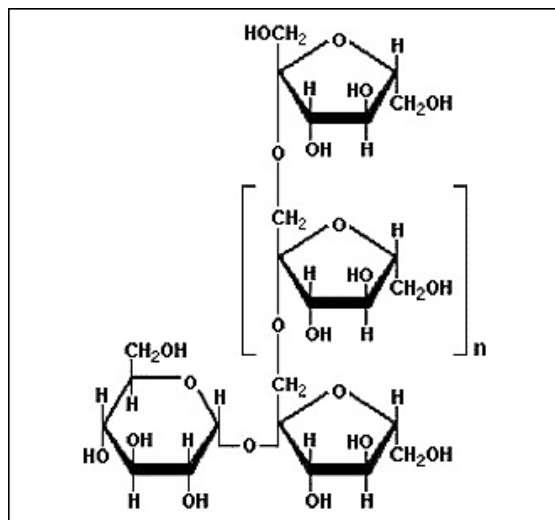
What is the maximum diameter of the largest food vacuole shown?

- A 8 μ m
 B 18 μ m
 C 180 μ m
 D 0.8 mm
- 2 Four different fruit juices, **A**, **B**, **C** and **D**, were tested with Benedict's solution. A second sample of each juice was hydrolysed and tested with Benedict's solution. The table shows the masses of the precipitates formed.

Which juice contains the greatest mass of non-reducing sugar?

	mass of precipitate before hydrolysis / mg	mass of precipitate after hydrolysis / mg
A	60	82
B	52	70
C	63	84
D	56	77

- 3 The diagram shows a molecule of inulin which is composed of a chain of 32 monomers.



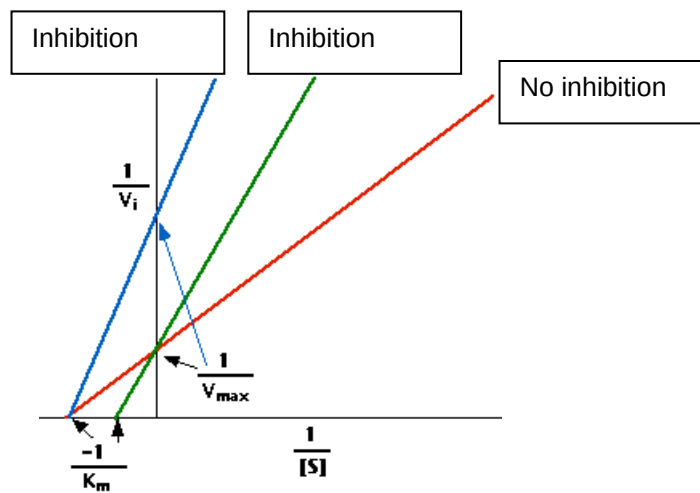
Which of the statements is correct?

- A The monomer of inulin is a five carbon sugar.
 - B The structure of inulin is highly branched.
 - C Inulin is more soluble than glucose.
 - D There is an α -(1,4) glycosidic bond in inulin.
- 4 Which are correct statements about triacylglycerols?
- 1 Esterification can occur in a step-wise manner to form non-polar triacylglycerols.
 - 2 Water is necessary to form polar triacylglycerol.
 - 3 H atom from glycerol and OH from fatty acids will combine to form water.
 - 4 Non polar hydroxyl groups from glycerol and polar carboxyl group from fatty acids form non-polar triacylglycerols.
 - 5 Polar hydroxyl groups from glycerol and polar carboxyl group from fatty acids form non-polar triacylglycerols.
- A 1, 2 and 3
 - B 2, 3 and 4
 - C 1, 3 and 5
 - D 1,3 and 4

5 Which set of statements correctly describes haemoglobin?

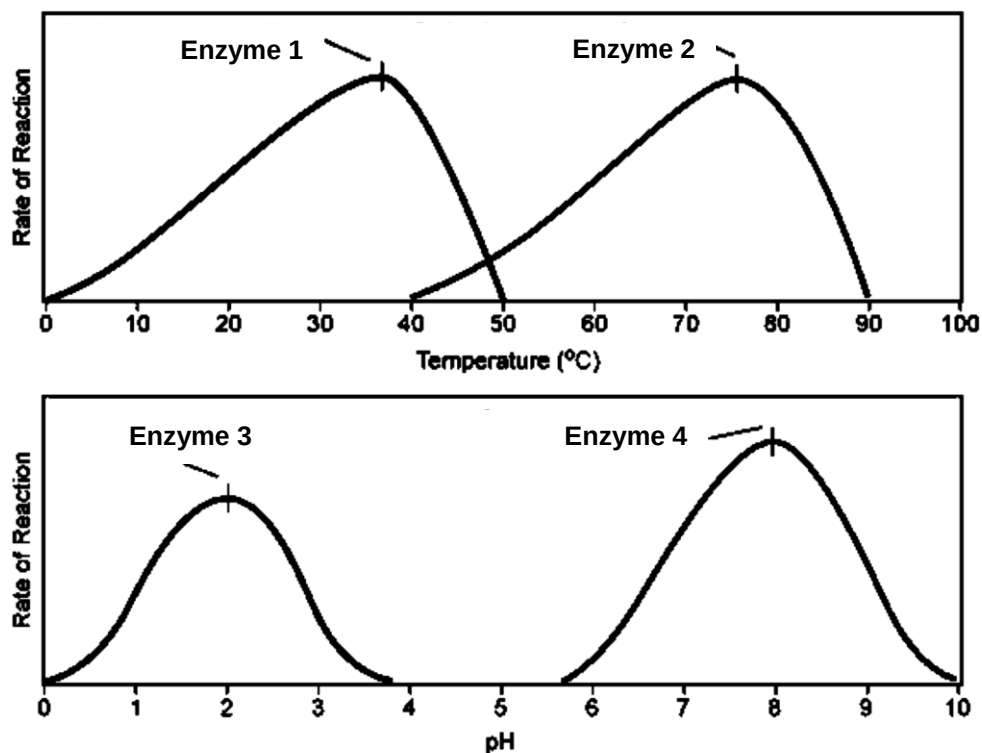
A	four polypeptide chains, each containing a prosthetic group	iron ions can associate with oxygen forming oxyhaemoglobin	in each chain, hydrophobic R groups of amino acids point towards the centre of the molecule	at 50% saturation, two oxygen molecules are transported by the molecule
B	four polypeptide chains, forming a globular chain	iron ions can associate with oxygen forming oxyhaemoglobin	in each chain, hydrophilic R groups of amino acids point towards the centre of the molecule	at 50% saturation, two oxygen molecules are transported by the molecule
C	four polypeptide chains, each containing a prosthetic group	iron ions in the molecule can bind reversibly with oxygen	in each chain, hydrophobic R groups of amino acids surround the iron ion	at 50% saturation, two oxygen atoms are transported by the molecule
D	polypeptide chains produce a loose helical shape, which curls to form a spherical molecule	iron ions can associate with oxygen forming oxyhaemoglobin	in each chain, hydrophobic R groups of amino acids point towards the centre of the molecule	at 50% saturation, two oxygen molecules are transported by the molecule

- 6 The diagram shows two types of inhibition, namely competitive and non-competitive inhibition. Which comparative statements about competitive and non-competitive inhibition are correct?



- 1 Competitive inhibitors compete with substrate and bind to active site while non-competitive inhibitors bind to substrate.
 - 2 In non-competitive inhibition, $\frac{1}{V_{max}}$ is changed with increasing substrate concentration while no change in competitive inhibition.
 - 3 In non-competitive inhibition, $\frac{1}{K_m}$ changes but in competitive inhibition, $\frac{1}{K_m}$ remains unchanged.
 - 4 In competitive inhibition, K_m increases while K_m is constant for non-competitive inhibition.
- A** 1 and 3
B 1, 2 and 3
C 1, 2 and 4
D 2 and 4

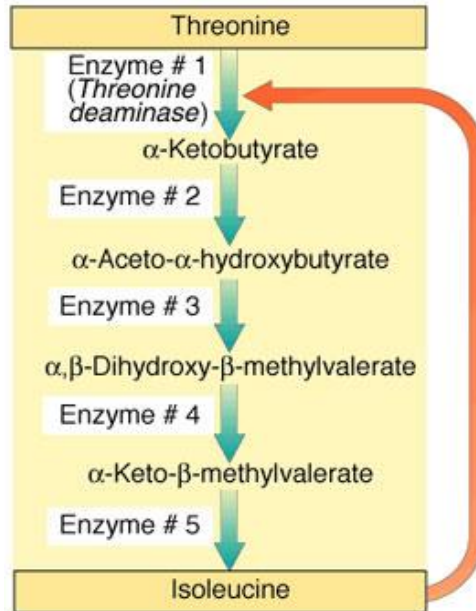
- 7 The graphs show the activities of 4 different enzymes under different conditions.



Which of the following statement is correct?

- A The kinetic energy of enzyme **2** and its substrate is fastest at 77°C.
- B At 80°C, several hydrogen bonds between the R groups of enzyme **2** are broken.
- C At pH 8, the substrate is bonded to the active site of enzyme **4** by hydrogen bonds only.
- D At pH 2, most of the R groups at the active site of enzyme **3** are negatively charged.

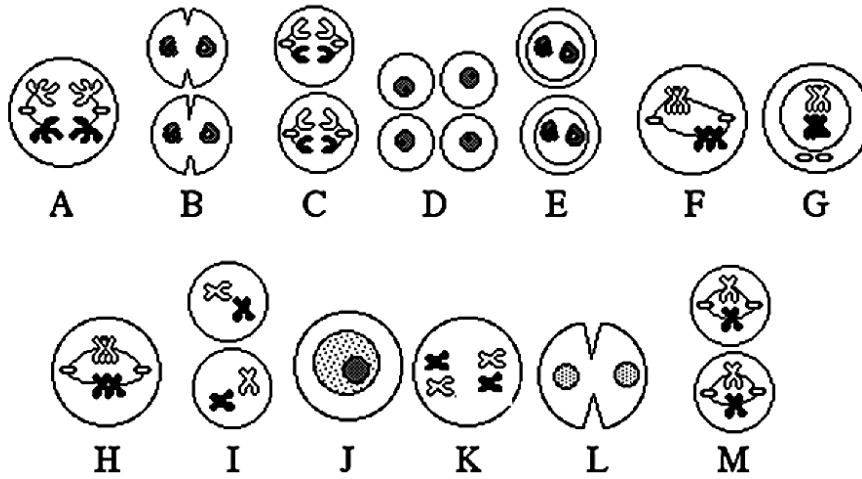
- 8 The synthetic pathway of isoleucine is outlined. When sufficient isoleucine has been produced, isoleucine binds to the enzyme threonine deaminase to stop the biosynthetic pathway.



Which of the following statements is true?

- A Threonine deaminase has at least 2 binding sites.
- B An increase in the amount of isoleucine decreases the amount of threonine produced.
- C Isoleucine is similar in shape and size to threonine.
- D There is a continuous production of isoleucine when Enzyme #1 is blocked.

- 9 The diagram below shows cell division in a diploid organism.

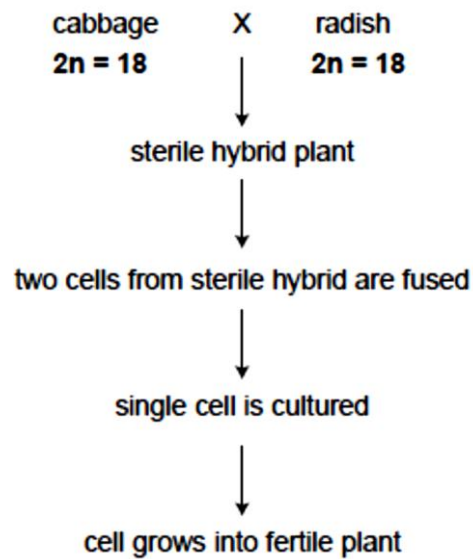


Which of the following statements are correct?

- I The cell at stage J is in G₂ phase.
- II Synapsis and crossing over occurs in the cell at stage K.
- III The order of sequence of events is J→G→K→H→F→A→L→I→M→C→E→B→D.
- IV The cell at stage F has the same number of chromosomes as a cell at stage C.

- A I and IV only
- B II and III only
- C I, II and IV
- D All of the above

- 10 Cabbage, *Brassica oleracea*, and radish, *Raphanus sativus*, are both members of the Brassicaceae family. The cells of each type of plant have 18 chromosomes. When these plants are crossed, a sterile hybrid is produced. Two cells from a sterile hybrid plant are fused and the single cell that is formed is cultured. This is summarised in the following diagram.

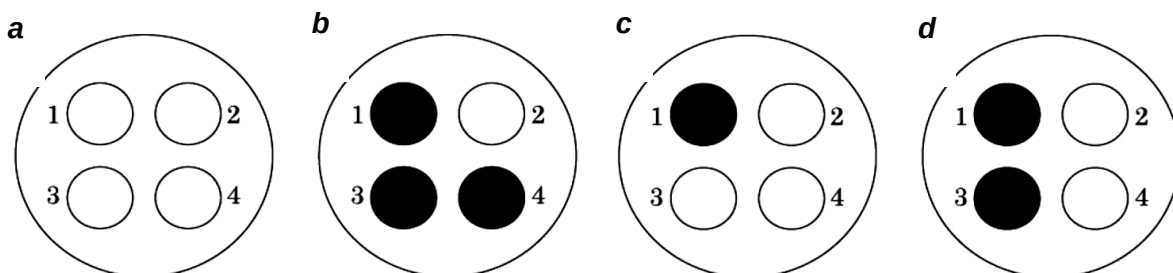


From this information, it is reasonable to conclude that

- A cabbage and radish have identical chromosomes.
- B gametes from the fertile plant will be identical with those of the cabbage or the radish.
- C the fused cell has 18 pairs of homologous chromosomes.
- D the sterile hybrid has nine pairs of homologous chromosomes.

- 11** Four mutant strains of bacteria (1 to 4) all require substance S to grow (each strain is blocked at one step in the S-biosynthesis pathway). Four plates were prepared with minimal medium and a trace of substance S, to allow a small amount of growth of mutant cells. On plate **a**, mutant cells of strain 1 were spread over the entire surface of the agar to form a thin lawn of bacteria. On plate **b**, the lawn was composed of mutant cells of strain 2, and so on.

On each plate, cells of each of the four mutant types were inoculated over the lawn, as indicated in the figure by the circles. Dark circles indicate excellent growth. A strain blocked at a later step in the S substance metabolic pathway accumulates intermediates that can 'feed' a strain blocked at an earlier step.



What is the order of genes (1 to 4) in the metabolic pathway for synthesis of substance S?

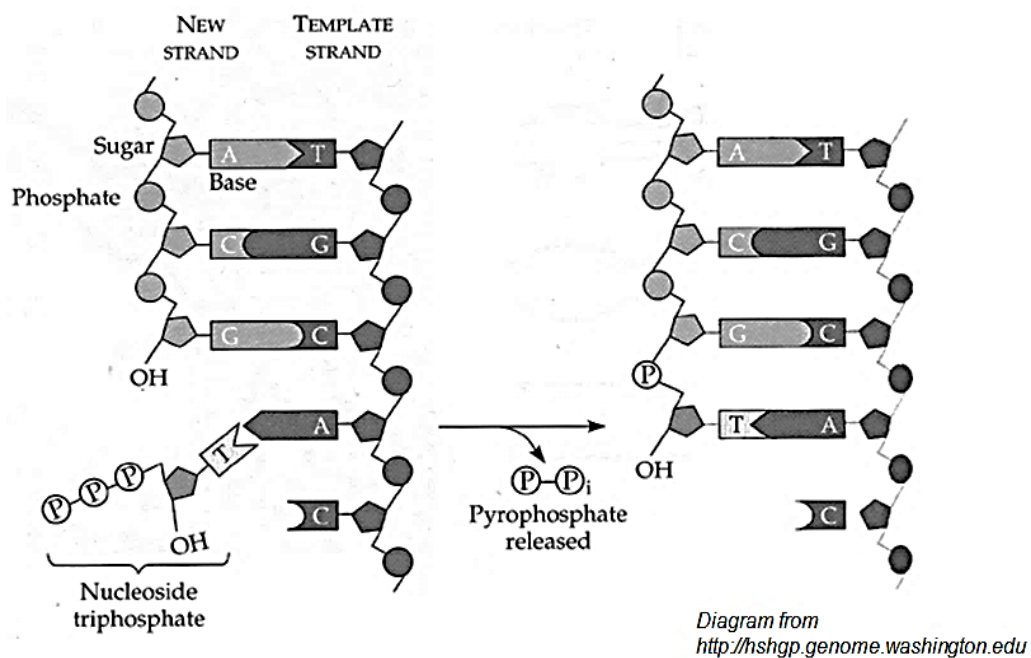
- A** $1 \rightarrow 2 \rightarrow 4 \rightarrow 3$
- B** $1 \rightarrow 3 \rightarrow 4 \rightarrow 2$
- C** $2 \rightarrow 1 \rightarrow 3 \rightarrow 4$
- D** $2 \rightarrow 4 \rightarrow 3 \rightarrow 1$
- 12** When DNA replicates, new nucleotides containing the common isotope of nitrogen (^{14}N) are used to build new nucleic acids.

In the laboratory, nucleotides can be synthesised using the heavy isotope of nitrogen (^{15}N). DNA is allowed to replicate once using these ^{15}N nucleotides, then two more times using ^{14}N nucleotides.

What will be the ratio of ^{14}N to ^{15}N nucleotides in the final molecules?

- A** 1 : 1
- B** 3 : 1
- C** 5 : 1
- D** 7 : 1

- 13 The diagram below shows the synthesis of a new strand of DNA during interphase.



Which of the following shows the correct combination of bond(s) that need to be formed and the kind of reaction that is involved in order for the nucleotide to be added to the DNA chain?

	Bond(s) to be formed	Reaction(s) involved
A	Phosphodiester	Condensation and Dephosphorylation
B	Phosphodiester	Condensation and Hydrolysis
C	Phosphoester and Hydrogen	Condensation and Hydrolysis
D	Phosphoester and Hydrogen	Condensation

- 14 Which of the following components are not directly required during translation?

- A Aminoacyl-tRNA synthetase
- B Small ribosomal subunit
- C Peptidyl transferase
- D rRNA

- 15** Acyclovir is a molecule that can substitute for the nitrogenous base, guanine. It is used as an antiviral drug.

Four features of acyclovir are listed below.

Which one of these features of acyclovir activity does not suggest that it could interfere with, and therefore inhibit, viral DNA replication?

- A** It can act as a competitive inhibitor of viral DNA polymerase.
- B** It can act as a non-competitive inhibitor of viral RNA polymerase.
- C** It can be incorporated into the growing viral DNA strand.
- D** It can be phosphorylated to become a nucleoside triphosphate.

- 16** The descriptions below are of nucleic acids in eukaryotes.

- 1 a polynucleotide of variable length formed by base pairing
- 2 a small polynucleotide with a specific three-dimensional shape
- 3 a small polynucleotide with specific sequence associated with proteins
- 4 a large polynucleotide with specific shape associated with proteins
- 5 a large polynucleotide with super-coiled sections associated with proteins

Which row correctly matches the description to its function?

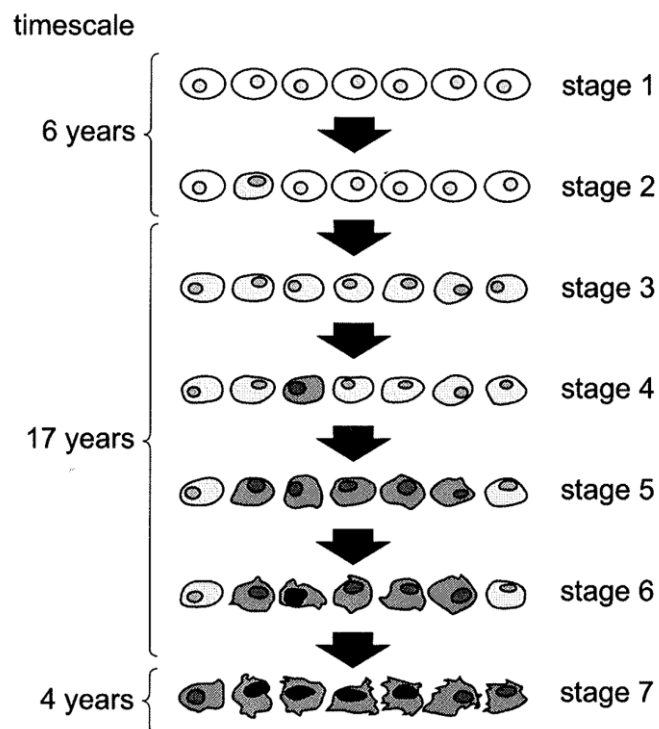
	carries specific amino acids	provides a site for protein synthesis	carries coded information	stores coded information	carries out transcriptional regulation
A	2	3	5	4	1
B	2	5	1	3	4
C	2	4	1	5	3
D	1	2	3	5	4

- 17** To fight influenza viral infections, which of the following strategies is valid?
- A** Inhibit the attachment and thereby entry of the virus into its host cell by developing antibodies that bind to the receptors on the viruses.
 - B** Inhibit the synthesis of the viral genome by developing an inhibitor that binds to DNA polymerase in the host cell.
 - C** Inhibit the expression of viral genes by developing RNA primers, which are identical to critical sections of the viral genomes.
 - D** Develop antibodies to neuraminidase to prevent entry into the host cells.
- 18** The human papillomavirus (HPV) may transform its target cell into a cervical cancer cell, which can lead to a tumour.

Which statement correctly explains this transformation?

- A** An oncogene codes for a protein that causes cell death.
- B** An oncogene codes for a protein that result in the cell dividing indefinitely.
- C** A proto-oncogene codes for a protein that causes cell death.
- D** A proto-oncogene codes for a protein that slows cell division.

- 19 The diagram represents stages in the development of a particular cancer in an adult person, over a period of 27 years.



Which of the following shows the processes happening between each of the stages indicated?

	Stage 1 to stage 2	Stage 3 to stage 4	Stage 4 to stage 5	Stage 5 to stage 6
A	mutation	proliferation	mutation and proliferation	mutation and proliferation
B	mutation	mutation	proliferation	mutation
C	mutation and proliferation	mutation and proliferation	proliferation	mutation
D	mutation	mutation and proliferation	proliferation	mutation and proliferation

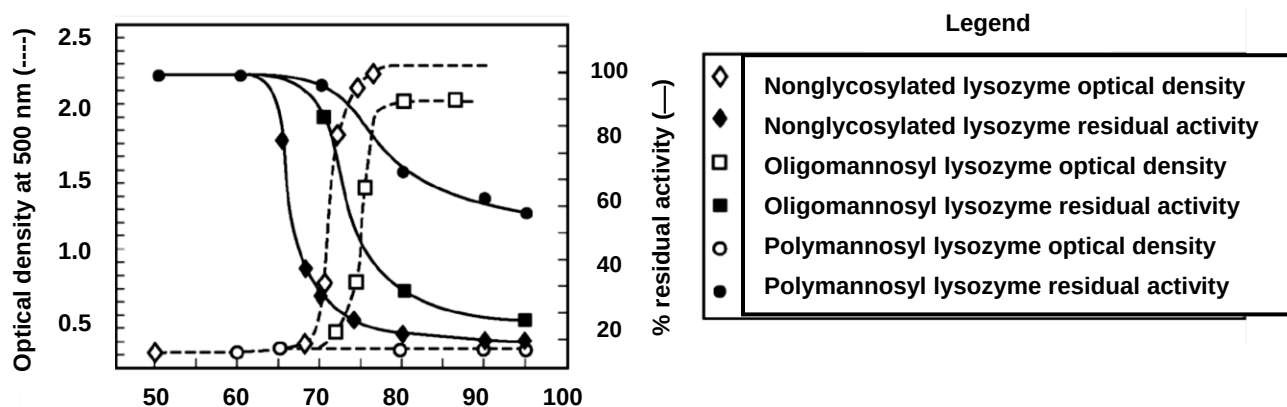
- 20 In a series of experiments designed to test the effect of glycosylation on proteins, the hen egg white enzyme lysozyme (MW ~14 kD) was genetically modified so as to introduce a site for N-glycosylation. When the protein was expressed in yeast, two forms of the enzyme were isolated: one with a short oligomannose chain, and one with a long polymannose chain. These proteins were examined for enzymatic activity with two different substrates: a soluble substrate, glycol chitin, and an insoluble substrate, a cell wall suspension from the bacterium *M. luteus*, as shown in the table below.

Further experiments were carried out in order to determine the thermal stability of the proteins, as shown in the figure below. Thermal stability was estimated by measuring the developed turbidity (precipitation) and residual enzymatic activity when samples were heated from 30 °C to 95 °C. Turbidity was measured as optical density at 500 nm. Residual enzyme activity was measured by hydrolysis of glycol chitin.

Enzymatic activity of glycosylated lysozymes

Lysozyme	Enzymatic activity (% of nonglycosylated enzyme)	
	Glycol chitin	<i>M. luteus</i> cell wall
Nonglycosylated	100	100
Oligomannosyl	100	73
Polymannosyl	81	11

Thermal stability of glycosylated lysozymes



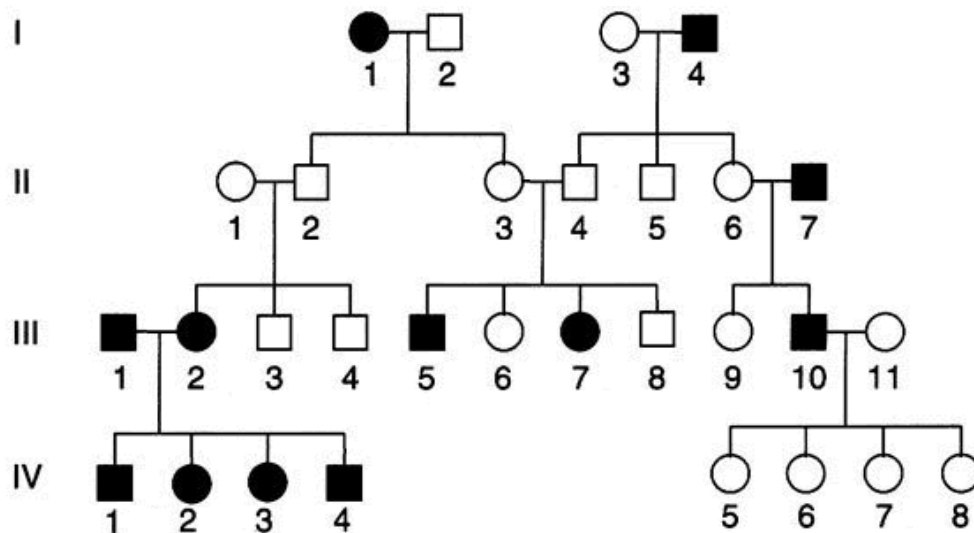
Which of the following most accurately describes the observed effects of glycosylation on thermal stability of the enzyme?

- A Glycosylation decreases the thermal stability.
- B Glycosylation with long chains decreases the thermal stability, but short chains have no effect.
- C Glycosylation increases the thermal stability and a greater increase occurs with long chains.
- D Glycosylation increases the thermal stability and a greater increase occurs with short chains.

- 21 A tall plant is crossed with a short plant, and the progeny are all intermediate in size between the two parental plants. The intermediate F_1 progeny were allowed to self-pollinate, and the F_2 progeny were also intermediate in size, but following a normal distribution. The results of the F_2 progeny is best explained by

- A environmental factors
- B multiple alleles
- C additive effect of polygenes
- D epistasis

- 22 In the pedigree chart below, the trait, adherent ear lobes, was studied.



The genotype(s) of III-3 and III-6, and the inheritance pattern are

	Genotype(s) of III-3 and III-6	Inheritance Pattern
A	AA or Aa	Sex-linked recessive
B	aa	Autosomal dominant
C	aa	Sex-linked dominant
D	AA or Aa	Autosomal recessive

- 23** In a genetic cross, a plant with genotype AaBb (F_1) was crossed with another plant with genotype aabb. In the F_2 progeny, the genotypes Aabb and aaBb were more frequent than that of AaBb and aabb. The genotypes of the pure-bred parental generation should be
- A** AaBb x AaBb
- B** AaBb x aaBb
- C** AAbb x aaBB
- D** AABB x aabb
- 24** A yellow flower, green-stemmed plant with the genotype YYrr was crossed with a white flower, red-stemmed plant with the genotype yyRR. The F_1 plants were allowed to self-fertilise. A chi-square test was carried out on the results obtained for the F_2 generation. Part of the table of values for chi-square is shown.

degrees of freedom	p = 0.5	p = 0.1	p = 0.05	p = 0.01	p = 0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.61	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.37	7.78	9.49	13.28	18.46
5	4.35	9.24	11.07	15.09	20.52

The value of chi-square in this investigation was 10.6.

What is the probability of this value and do the results fit the expected ratio?

	Probability	Results fit expected ratio
A	Between 0.01 and 0.05	No
B	Between 0.01 and 0.05	Yes
C	Between 0.05 and 0.1	Yes
D	Between 0.1 and 0.5	No

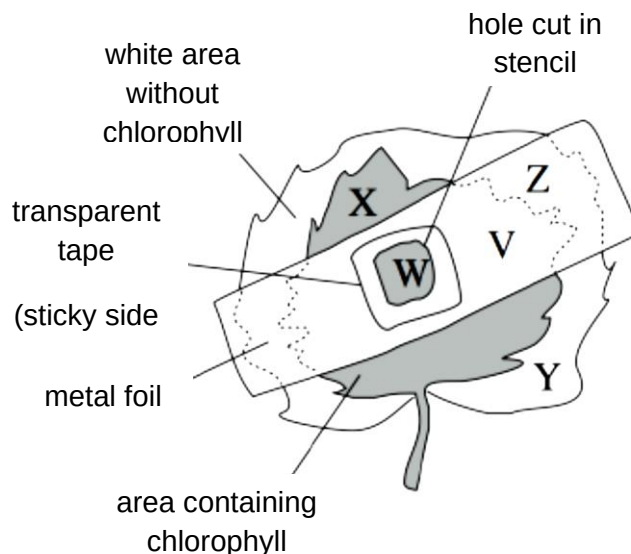
- 25 In a series of breeding experiments using *Drosophila*, a linkage group composed of five genes was found to show the following recombination frequencies per 100 fertilised eggs.

	Single bristles (<i>sn</i>)	Vermilion eye (<i>v</i>)	White eye (<i>w</i>)	Ruby eye (<i>rb</i>)	Tan body (<i>t</i>)
Single bristles (<i>sn</i>)	0.0	12.0	19.5	13.5	6.5
Vermilion eye (<i>v</i>)	12.0	0.0	31.5	25.5	5.5
White eye (<i>w</i>)	19.5	31.5	0.0	6.0	26.0
Ruby eye (<i>rb</i>)	13.5	25.5	6.0	0.0	20.0
Tan body (<i>t</i>)	6.5	5.5	26.0	20.0	0.0

From this table of results, it can be deduced that the correct sequence of the genes on the chromosome is

- A** $sn - t - v - rb - w$
- B** $sn - v - w - rb - t$
- C** $w - rb - sn - t - v$
- D** $w - rb - t - sn - v$

- 26 In an experiment to test the hypothesis that leaves need both light and chlorophyll to make starch, a stencil was fixed to both sides of a leaf of a variegated geranium plant as shown.



The hole cut in the stencil was covered by a small piece of transparent tape. As a result of this treatment, the leaf had been divided into five parts experiencing different conditions, the positions of which are indicated by the letters V – Z in the diagram above.

V contained chlorophyll but was in darkness.

W contained chlorophyll and was open to the light but was covered with transparent tape.

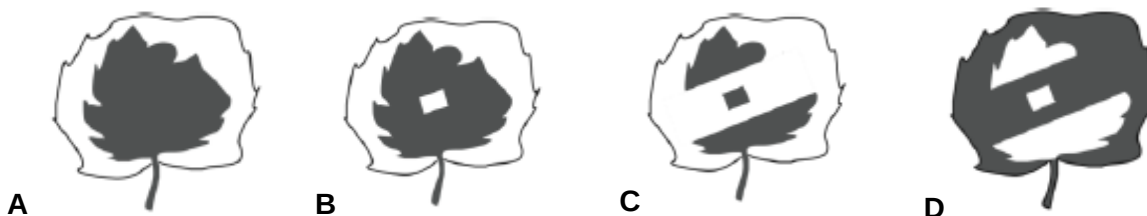
X contained chlorophyll and was open to the light but not covered with transparent tape.

Y had no chlorophyll and was open to the light.

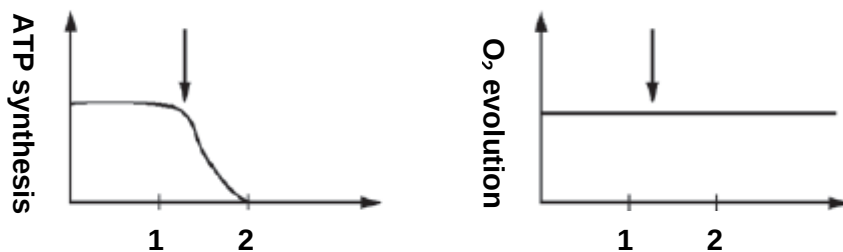
Z had no chlorophyll and was in darkness.

The plant was then left in bright light for 48 hours, after which the stencil was removed and the leaf tested for starch.

Which of the diagrams A - D would indicate the appearance of the leaf after testing for starch? **Note: Grey area = test positive for starch**



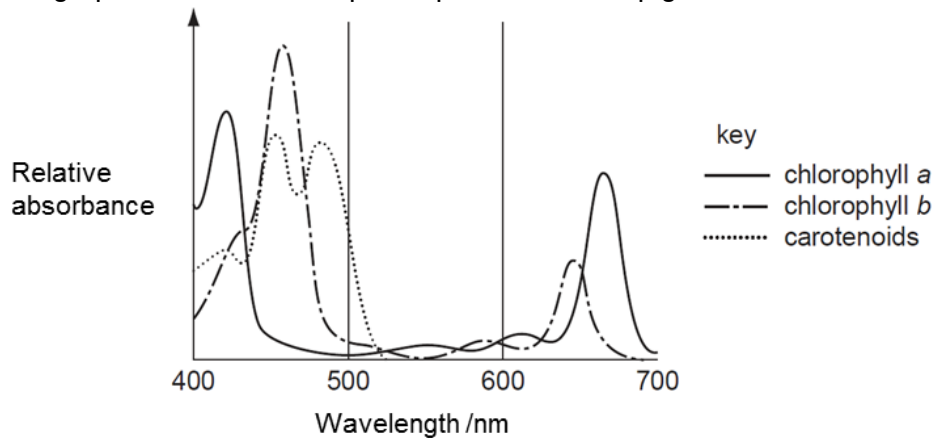
- 27 A chemical compound, A, has properties that suggest it would interfere with chloroplast function. The effect of the compound has been tested using an illuminated suspension of isolated chloroplasts and measuring the rates of ATP synthesis and O_2 production. The result is shown in the graphs below with the arrow indicating the time of addition of compound A.



Which of the following statements is true for the chloroplasts treated with compound A?

- A No proton gradient exists across the thylakoid membrane, but the reduction of $NADP^+$ increases to a new, higher steady-state level.
- B No proton gradient exists across the thylakoid membrane, but the reduction of $NADP^+$ continues at the initial rate.
- C No proton gradient exists across the thylakoid membrane and the reduction of $NADP^+$ drops to zero.
- D The proton gradient across the thylakoid membrane is unaffected, but the reduction of $NADP^+$ drops to zero.

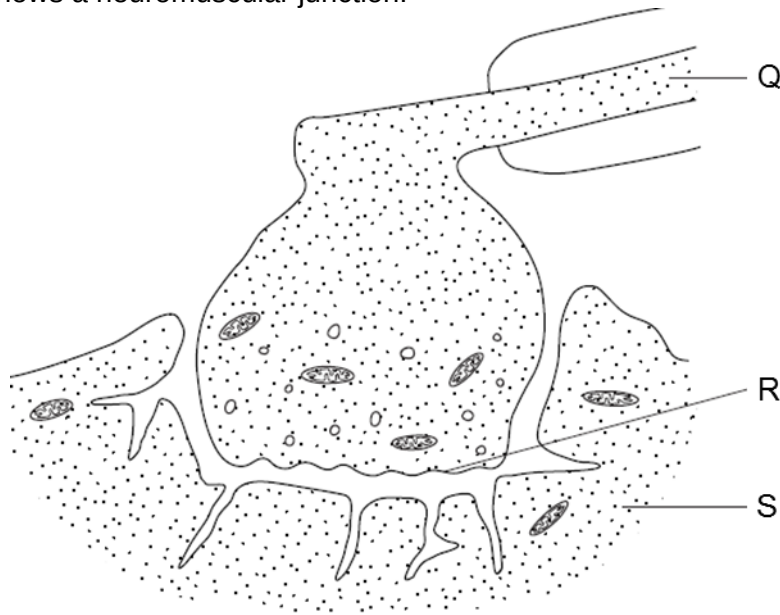
- 28 The graph shows the absorption spectra of some pigments found in chloroplasts.



Which statement is **not** correct?

- A Photosynthesis will be fastest in light at the red end of the spectrum, as red light has higher energy than blue light.
 - B Most leaves are green as chlorophyll absorbs light in the blue and red regions of the spectrum.
 - C Prior to leaf fall, chlorophyll is broken down, leaving carotenoids which makes leaves look yellow or red.
 - D Having several pigments, rather than one, increases the efficiency of photosynthesis.
- 29 Which of the following correctly describes how a mammal maintains a constant internal environment?
- A When there is a deviation from reference point, the detector detects the deviation and transmits the information to the control centre.
 - B The control centre carries out the response to cancel the deviation, causing a fall or rise back to reference point.
 - C The effector receives input from the detector and compares the input with the reference point.
 - D When reference point is reached, there is feedback information and the mechanism is triggered to enhance the deviation in the opposite direction from reference point.

30 The diagram shows a neuromuscular junction.



Which row shows the correct structure to function relationship?

	Q	R	S
A	axon allows for transmission of electrical impulse	post-synaptic membrane allows for exocytosis of neurotransmitters	muscle cell will undergo contraction upon receiving chemical signals
B	axon allows for movement of neurotransmitters from cell body to the post-synaptic membrane	pre-synaptic membrane allows for exocytosis of neurotransmitters	signal transduction will occur in cytoplasm of muscle cell upon activation of chemically-gated receptors
C	axon allows for transmission of electrical impulse	pre-synaptic membrane allows for exocytosis of neurotransmitters	signal transduction will occur in cytoplasm of muscle cell upon activation of chemically-gated receptors
D	axon allows for movement of neurotransmitters from cell body to the pre-synaptic membrane	post-synaptic membrane allows for exocytosis of neurotransmitters	muscle cell will undergo contraction upon receiving chemical signals

- 31** In one taxonomic classification, Achaea, Eukarya, and Bacteria represent the three major domains of life. Eukarya utilise the general transcription factors TATA-binding protein (TBP) and TFIIB in transcription, whereas Bacteria do not. At least one member of Achaea has a protein similar to TBP and a protein similar to TFIIB. Based on this observation, which of the following scenarios is most likely?
- A** Achaea and Eukarya diverged after their common ancestor diverged from Bacteria.
 - B** Achaea and Bacteria diverged after their common ancestor diverged from Eukarya.
 - C** Achaea, Bacteria and Eukarya diverged simultaneously from a common ancestor.
 - D** Bacteria and Eukarya diverged after their common ancestor diverged from Achaea.
- 32** Which of the following best supports the endosymbiotic theory of the evolutionary origin of mitochondria?
- A** Mitochondria, chloroplasts, and prokaryotes contain electron carriers.
 - B** Genes for mitochondrial pyruvate dehydrogenase subunits are found in the nuclear DNA.
 - C** Many mitochondrial proteins are imported across both inner and outer membranes after translation on cytoplasmic ribosomes is completed.
 - D** Mitochondrial and bacterial ribosomal functions are inhibited by the same antibiotics.

- 33** The mussel *Mytilus edulis* thrives in saline habitats, in both the highly salty seawater of tidal zones and the less salty estuaries. This results in two kinds of populations: one adapted to the higher salt concentrations of the tidal zone, and one adapted to the lower salt concentrations of the estuary.

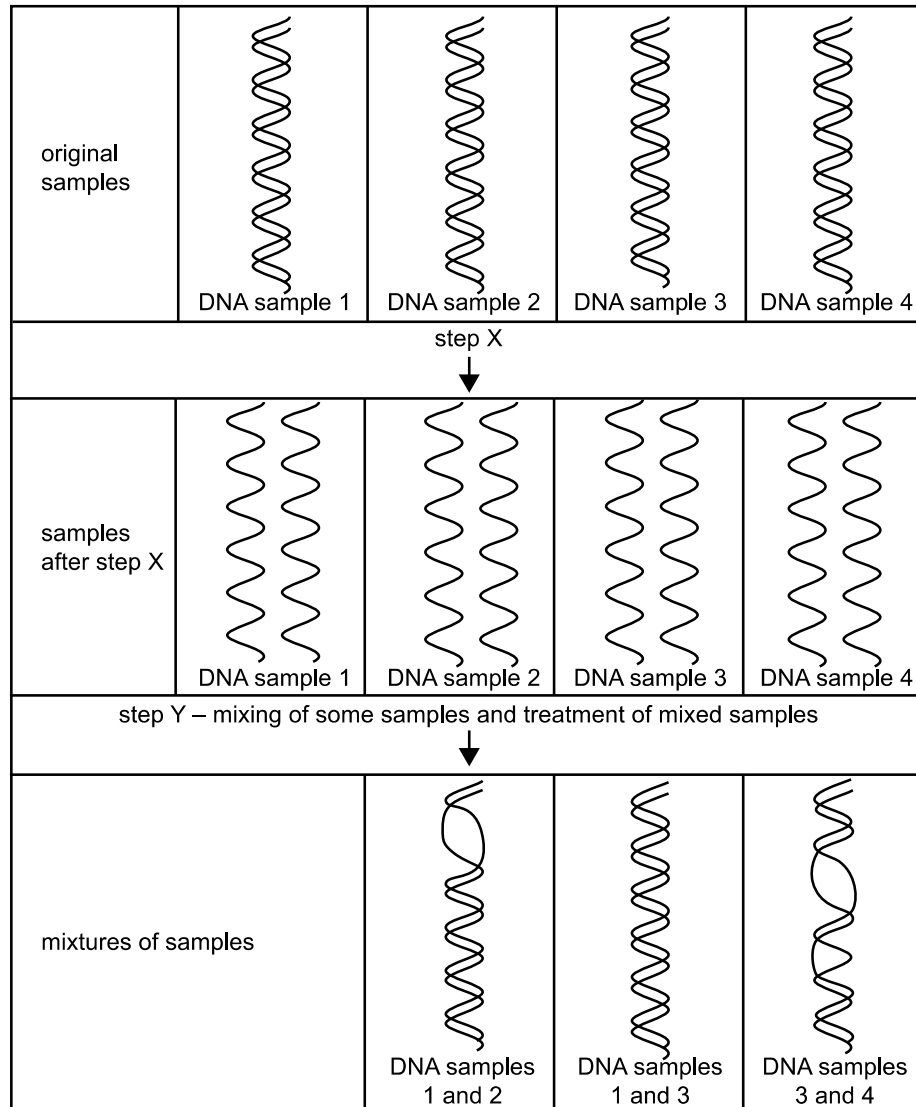
It has been found that the more salt-tolerant populations have high frequencies of an allele that produces an enzyme involved in maintaining osmotic equilibrium. Conversely, estuarine mussels having the same enzyme seem to be disfavoured and have a much higher death rate than mussels without the allele. Adult estuarine populations do have lower frequencies of this allele.

Each spring, large numbers of larvae from the salty habitats pour into the estuaries.

The invasion of the seawater larvae would be expected to facilitate change in the gene pool of the estuarine population by a process called

- A** directional selection.
- B** disruptive selection.
- C** genetic drift.
- D** gene flow.

- 34** Samples of DNA were taken from four individuals. The samples went through a series of steps and the resulting DNA is shown.



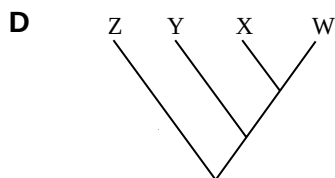
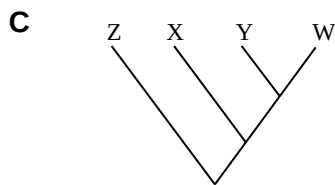
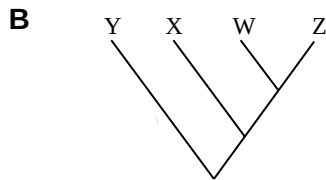
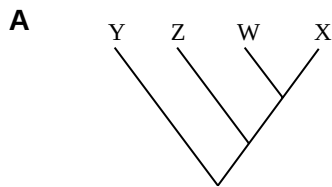
Using information from the diagram, which conclusion can be drawn?

- A** Individuals belonging to samples 1 and 2 are less closely related than individuals belonging to samples 3 and 4.
- B** Individuals belonging to samples 1 and 3 are more closely related than individuals belonging to samples 1 and 2.
- C** Samples 1 and 2 are from individuals belonging to the same species.
- D** Treatment at step X would have included cooling each of the original samples.

- 35** A cladogram is a branching diagram in which groups of closely related species (i.e. those sharing many characteristics) are shown to have branched away from the common line of descent via the same ancestor. The table gives a number of characteristics which are found in a group of four species, W, X, Y and Z. Shading indicates the characteristic is present.

Characteristic	Species			
	W	X	Y	Z
Fingers and toes				
Endothermy				
3 ear ossicles				
Amnion				
Placenta				
Internal fertilisation				
Mammary glands				
Oviparity (lays eggs)				
Webbed feet				
Hair				
Feathers				

Which one of the cladograms below is the most likely hypothesis explaining the distribution of characters?



- 36 When a polymerase chain reaction (PCR) amplification was performed on human genomic DNA, multiple products of varying sizes were obtained, including one of the expected size. Which of the following modifications to the protocol is most likely to eliminate the extra PCR products?
- A Raising the annealing temperature from 52 °C to 56 °C
 - B Raising the denaturation temperature from 94 °C to 96 °C
 - C Raising the elongation temperature from 70 °C to 74 °C
 - D Increasing the elongation time from 3 minutes to 4 minutes
- 37 *Bam*HI is a restriction enzyme produced by *Bacillus amyloliquefaciens*. It recognises the specific nucleotide sequence GGATCC. *B. amyloliquefaciens* is susceptible to attack by viruses known as bacteriophages.

Which statement describes the natural function of *Bam*HI?

- A *Bam*HI enzymatic modification of nucleotides occurs in all the GGATCC nucleotide sequences of *B. amyloliquefaciens*, to protect DNA from degradation by bacteriophage enzymes and prevent the nucleotides from being used for viral nucleic acid synthesis.
- B Infectivity rate is lowered when *B. amyloliquefaciens* is subjected to invasion by bacteriophages that have double-stranded DNA as their genetic material, since *Bam*HI can recognise and cleave specific viral nucleotide sequences.
- C Restriction fragments produced as a result of *Bam*HI binding to and cleaving DNA at all sites within the bacterial cell that have a GGATCC nucleotide sequence, prevent integration of bacteriophage nucleic acid into the bacterial genome.
- D Transcription and translation of the *B. amyloliquefaciens* gene encoding the *Bam*HI enzyme is followed by secretion of the enzyme into the external environment, where it has antibacterial action against non-related bacterial species.

38 Gene therapy is used to treat inherited diseases such as cystic fibrosis. Some of the concerns about gene therapy are listed below.

- 1 Most gene therapy must be repeated in succeeding generations since germ cells are not involved.
- 2 Genetically modified organisms used in producing the gene therapy may escape into the environment with unforeseen consequences.
- 3 Putting genes into the germ line affects subsequent generations and is banned in many countries.
- 4 The same techniques for treating serious, life-threatening conditions may be used to try to change other things such as intelligence and skin colour.
- 5 Viral vectors, such as those used in the treatment of cystic fibrosis, have been known to produce harmful side effects.

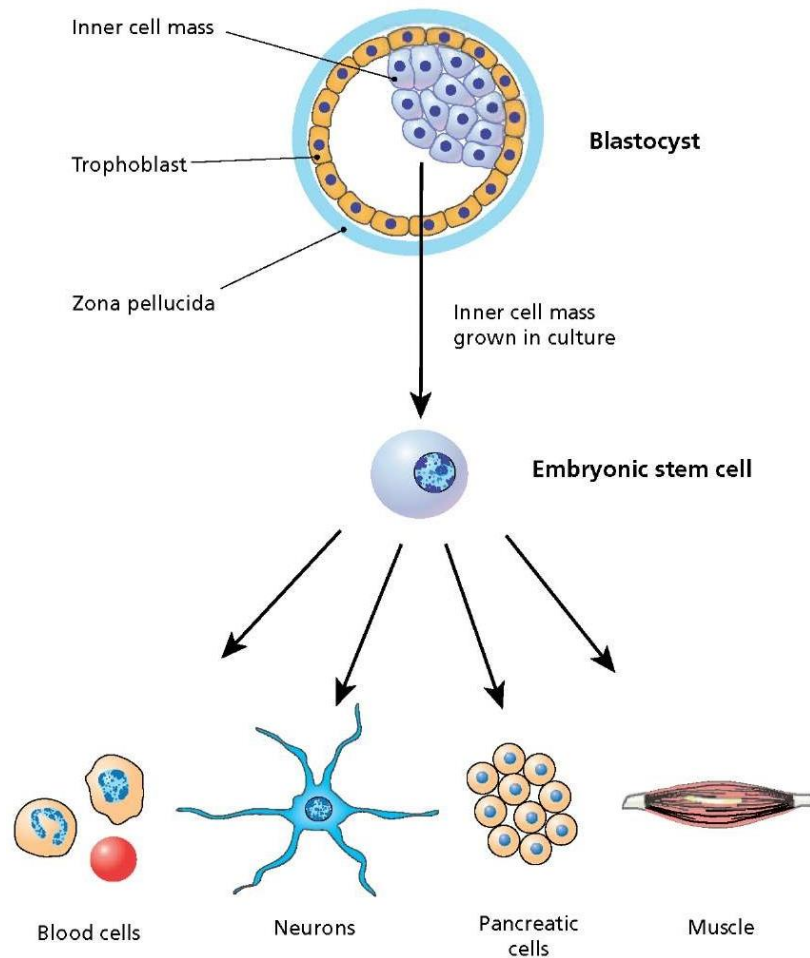
Which row identifies the ethical concerns?

- A** 3, 4 and 5
- B** 1, 3 and 5
- C** 2, 3 and 4
- D** 1 and 2

39 Which of the following is valid ethical concern on the use of stem cells?

- A** Use of embryonic stem cells could result in the formation of new organism.
- B** Totipotent stem cells should be recommended as they are in huge abundance.
- C** Skin stem cells usage should be prohibited as they are limited in quantities.
- D** Embryonic stem cells retrieval results in the destruction of human embryos.

- 40 Which of the following features of the embryonic stem cells and specialized cells shown in the diagram are true?



	Embryonic stem cells	Specialized cells
A	Embryonic stem cells display greater plasticity when grown in culture than when in blastocyst.	The blood cells are genetically different from the embryonic stem cells but have shorter telomeres.
B	Embryonic stem cells are pluripotent and are capable of transdifferentiating into many different cell types.	The pancreatic cells are genetically identical to the embryonic stem cells but with a different set of genes expressed.
C	Embryonic stem cells are multipotent and are capable of differentiating into limited range of cell types.	The blood cells are genetically different from the embryonic stem cells because different genes are expressed.
D	Embryonic stem cells are pluripotent and are capable of differentiating into many different cells types.	The pancreatic cells are genetically identical to the embryonic stem cells but have shorter telomeres.

2015 H2 Biology Preliminary Examination Paper 1 (Answer key)

QN	ANS	QN	ANS	QN	ANS	QN	ANS
1	A	11	B	21	C	31	A
2	A	12	D	22	D	32	D
3	B	13	D	23	C	33	D
4	C	14	A	24	A	34	B
5	A	15	B	25	C	35	D
6	D	16	C	26	C	36	A
7	B	17	A	27	B	37	B
8	A	18	B	28	A	38	C
9	A	19	B	29	A	39	D
10	C	20	C	30	C	40	D